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An Ultrasound-based Localization Algorithm for Catheter Ablation Guidance in the Left Atrium

Abstract

We present a method for catheter localization in the left atrium based on the unscented particle filter (UPF), a Monte Carlo method employed in stochastic state estimation. Using an intracardiac echo (ICE) ultrasound catheter, we acquire ultrasound images of the atrium from multiple configurations and iteratively determine the catheter's pose with respect to anatomy. At each time step, the catheter's change in pose is determined using either a six-degree-of-freedom electromagnetic pose sensor or a robotic guide catheter

whose kinematics serve as a pseudo-pose measurement. Sensor and kinematic model uncertainties are explicitly considered when computing the localization estimate. Acquired ultrasound images are compared with simulated ultrasound images based on segmented computed tomography (CT) or magnetic resonance (MR) data of the left atrium. The results of these comparisons are used to refine the localization estimate. After considering less than 30 seconds' worth of ICE data, our algorithm converges to an accurate pose estimate. Furthermore, our algorithm is robust to sensor drift and kinematic model errors, as well as gradual, unmodeled movements in the anatomy. Such problems typically complicate traditional image-based localization.

The International Journal of Robotics Research

Vol. 29, No. 6, May 2010, pp. 643–665

DOI: 10.1177/0278364909105332

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Figures 1, 3–23 appear in color online: <http://ijr.sagepub.com>

KEY WORDS—medical robots and systems; human-centered and lifelike robotics; localization; mobile and distributed robotics SLAM; sensor fusion; sensing and perception computer vision.

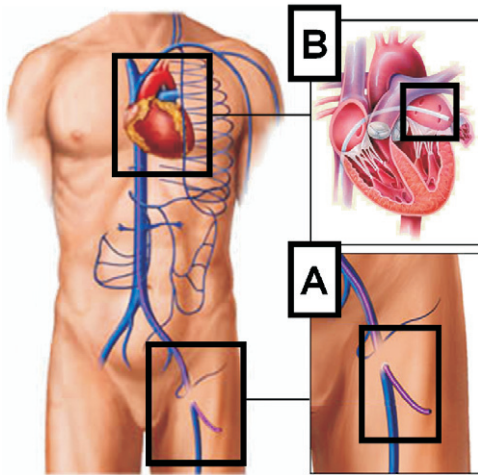


Fig. 1. Catheter ablation procedure. AF can be treated through cardiac tissue ablation. An ablation catheter is inserted through the femoral vein (A) and its tip emerges inside the heart. The tip is maneuvered around the left atrium (B) using pull-wires running down the catheter's body.

1. Introduction

Over the last 30 years, the trend in surgery has been to operate on patients in a decreasingly invasive fashion. Several procedures that once required substantial incisions to access internal organs are now performed through miniature “port-like” incisions. The advantages of operating using minimally invasive surgery (MIS) techniques include reduced risks of infection and expedited patient recovery. However, technological limitations have relegated MIS to procedures that are either relatively simple, or for which some special-purpose device can be designed.

The treatment of atrial fibrillation (AF) by catheter ablation exemplifies the successes and shortcomings of MIS. AF is a widely prevalent cardiac arrhythmia, and a cause of significant morbidity, mortality and health care expenditure. The Cox Maze procedure, a commonly used technique in the treatment of AF, interrupts aberrant conduction pathways by ablating (burning) tissue in the left atrium (Cox et al. 1996). As the Cox Maze was initially performed as open-heart surgery, its role was limited to that of an adjunct operation applied in the setting of other open-heart procedures.

Haïssaguerre et al. (1998) reported success in the MIS treatment of atrial fibrillation using a catheter with an ablation device attached at its tip (Figure 1); since then, MIS treatment of atrial fibrillation by means of catheter ablation has become increasingly common.

Unfortunately, difficulties with catheter manipulation and inadequate facilities for visualizing the catheter inside the anatomy have resulted in lengthy procedure times of the order of several hours, and limited success rates: less than 75%

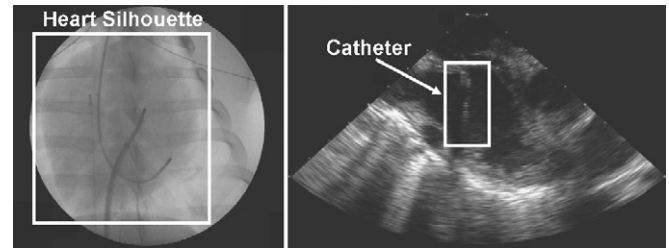


Fig. 2. Imaging insufficiencies. X-ray (left) does not yield good definition of the cardiac structure while cardiac ultrasound (right) presents complications for catheter imaging.

according to recent patient studies (Mansour et al. 2004; Nanthakumar et al. 2004). While ablation catheters can be steered to reach virtually every part of the left atrium, smooth, continuous motions for creating contiguous ablation lesions is challenging.

Complicating matters further, traditional imaging modalities used for guiding the catheter ablation procedure fail to yield a clear and comprehensive picture of the surgical environment (Figure 2). For example, X-ray fluoroscopy images the ablation catheter with clarity, but delivers inadequate contrast for imaging cardiac tissue. Conversely, intracardiac echo (ICE) ultrasound readily images cardiac tissue, but is difficult to orient for imaging the ablation catheter tip.

Furthermore, both X-ray and ICE yield two-dimensional (2D) images, requiring the physician to mentally piece together images taken from different views to recover three-dimensional (3D) information. As a consequence of poor visual feedback, navigating the catheter tip to ablation sites often requires several attempts before precise positioning and tissue contact are achieved.

2. Contributions

In this paper we address the problem of visualizing the catheter inside the anatomy by continuously estimating the *pose* (position and orientation) of the catheter with respect to the left atrium. We display our estimate as a 3D rendering of a “virtual catheter” inside an anatomically representative surface mesh model (Figure 3).

The advantage of this rendered view over X-ray and ICE images is that both the catheter and the anatomy are displayed clearly in three dimensions, with facilities for observing the data from different vantage points without repositioning imaging hardware. While determining the catheter's pose within the anatomy is one of several factors limiting performance in the catheter ablation procedure, we believe that doing so can help mitigate the effects of other limiting factors, including the time required for electrical signal mapping and catheter steering.

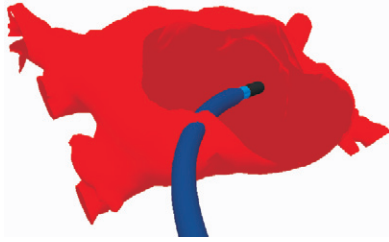


Fig. 3. Virtual catheter display. We estimate the pose of the ablation catheter with respect to the left atrium, and display our estimate as a virtual rendering of the catheter and anatomy.

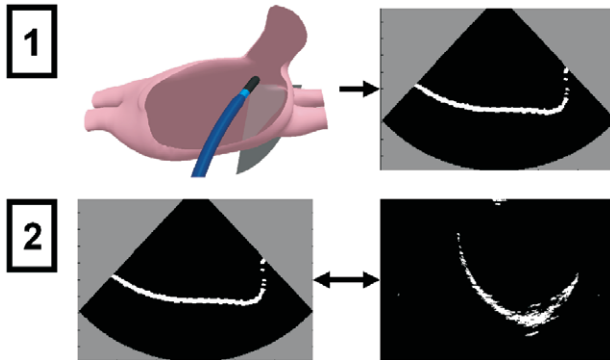


Fig. 4. Pose estimation algorithm. In (a) we generate a “virtual ICE” image based on a particular catheter pose estimate (1). The gray wedge emanating from the catheter tip represents the region imaged under ultrasound. The virtual ICE image is compared with the acquired ICE image (2), yielding a correlation score.

We propose a novel solution for catheter pose estimation in the left atrium based on Monte Carlo localization (MCL), a technique for determining the position of a *mobile robot* within a *map* of its *environment*. Here, our *mobile robot* is an ICE catheter whose images serve as range data of the left atrium *environment*. The *map* we localize with respect to is an atrial surface mesh segmented from pre-operative computed tomography (CT). We initially described this technique in Koolwal et al. (2008).

We compare acquired ICE images to “virtual ICE” images generated from different ICE catheter pose estimates (Figure 4). Pose estimates whose virtual images do not match acquired images are discarded and, over time, only accurate estimates remain in consideration.

As the ICE catheter moves inside the left atrium, an electromagnetic position sensor (EPS) mounted on its tip informs our algorithm of its change in pose at each time step. EPS measurements are considered differentially to negate accumulated sensor drift. Once the ICE catheter is localized with respect

to the anatomy, we compute the registration parameters linking EPS measurements to surface mesh coordinates. Finally, we localize the ablation catheter inside the surface mesh using measurements from an EPS sensor mounted on the ablation catheter’s tip.

Our algorithm can also function without EPS when the ICE catheter is steered using a robotically controlled guide catheter. This approach would be useful in a scenario where an ultrasound imaging transducer is integrated onto the ablation catheter (Gentry and Smith 2004), or if the ablation catheter is localized with respect to the ICE catheter using a segmentation-based tracking approach (Linguraru et al. 2007; Novotny et al. 2007). Without EPS sensors, we must rely on the robotic catheter’s kinematics to serve as a proxy measurement for the change in ICE catheter pose. While the kinematic model may be less accurate than EPS measurements, it is not subject to errors induced by reference electrode movement, and precludes the need to set up, operate, and monitor sensors during the procedure. To compensate for kinematic model inaccuracies, we devise a method for incorporating kinematic model uncertainty into our estimation scheme.

We evaluate our algorithm *in vitro* using a left atrium phantom, described in Section 4.3.1. Our study can be considered a proof of concept, with the understanding that there are several details to address before employing our algorithms *in vivo*. For example, the shape of the atrium typically deforms between pre-operative and intra-operative imaging, implying that a surface mesh generated from a pre-operative CT scan may be “out of date” at the time of operation. We do not attempt to address this issue in this paper, but will address it in subsequent work, where we use acquired ICE images to change the pre-operative model during the localization process. Further, we plan to evaluate our algorithms in a porcine model once facilities for doing so become available.

3. Related Work

In the catheter ablation procedure, ablation lines are “drawn” over tissue surfaces in the left atrium. Of particular interest are the posterior wall, the pulmonary vein (PV) openings, and the atrium’s roof (O’Neill et al. 2007). While the longitudinal distance across the atrium can be as large as 80 mm, the PV openings on either side are on the order of 10 mm in diameter (Lickfett et al. 2005). As such, not only is it important to visualize the entire atrium for aiding gross catheter movements, it is also important to have a high-resolution exploded view of the PVs to aid in precise tracing of circular ablation lines around the PV openings.

Unlike X-ray fluoroscopy and ICE, magnetic resonance (MR) and CT yield high-resolution 3D images that can readily be segmented to isolate tissue structures. Before undergoing the catheter ablation procedure, patients often receive a cardiac CT scan from which the left atrium is segmented. A

surface mesh reconstruction of the segmented atrium allows physicians to easily visualize a patient's specific anatomy in three dimensions. This mesh would be ideal for catheter ablation guidance, but traditional MR and CT imaging systems can only be used in a pre-operative setting. Furthermore, MR and CT require time for image recovery, segmentation, and surface mesh reconstruction that would preclude intra-operative imaging.

Recent advancements have enabled real-time MR imaging over small volumes of interest, such as the region of space immediately surrounding the ablation catheter tip (Omary et al. 2003; Horvath et al. 2007; Nazarian et al. 2008). However, ferrous metals inside the operating room can negatively affect the fields generated by MR magnets, not only resulting in distorted images, but endangering the patient as well (Nazarian et al. 2008). Real-time CT can be approximated intra-operatively using rotational X-ray, whereby a series of 2D X-ray images are rapidly acquired from different angles and a 3D volume is reconstructed from the data (Manzke et al. 2006, 2007; Thiagalingam et al. 2008). The major downside of this approach is the large dose of radiation administered to the patient for each generated volume (Manzke et al. 2006). Finally, 3D ultrasound (3DUS) has gained popularity because it is relatively cheap and harmless to the patient. Unfortunately, cardiac 3DUS suffers from a limited field of view and is difficult to segment for surface mesh reconstruction and catheter tip identification (Cannon et al. 2003; Zhang et al. 2004; Linguraru et al. 2007; Novotny et al. 2007). As a result, 3DUS data is generally volume-rendered for visual presentation, but structures are not as clearly identifiable as they are in a properly segmented surface mesh. Another hindrance to its acceptance, cardiac 3DUS presently requires patient intubation for probe access and image acquisition, a risk physicians prefer not to take unless explicitly required.

The lack of adequate imaging techniques for catheter guidance has led to a variety of sensor-based visualization techniques. Perhaps the most widely adopted involves the use of EPS sensors that record the location of the catheter's tip in space. Sun et al. (2007) and Wein et al. (2007) used an EPS sensor to track the tip of an ICE catheter, allowing intra-operative ICE data to be registered with pre-operative CT data. Wein et al. (2007) enhanced this process by simulating ICE images using the CT data; acquired images are then compared with simulated images for more accurate correlation computations. Both techniques require an initial manual registration, followed by an automated optimization process.

Clinically, a more widespread solution involves generating a 3D model that resembles the shape of the atria by computing the hull of all recorded tip positions; physicians can then use this "point cloud model" as an anatomic reference for guidance. During the catheter ablation procedure, one of two different EPS technologies, Ensite NavX (St Jude Medical, St Paul, MN) or Carto XP/CartoMerge (Bio-Sense Webster, Diamond Bar, CA), are commonly employed to "map out" an atrial point

cloud model. Both rely on relative measurements taken between the ablation catheter and reference electrodes placed on or inside the body. These reference electrodes are assumed to remain stationary during the procedure.

An extension to EPS mapping is to register the resultant point cloud model with an atrial surface mesh based on segmented CT or MR. After the initial registration is computed, subsequent recorded tip positions are displayed inside the mesh, which has greater structural detail than the point cloud model. The registration technique generally used is a variant of the iterative closest point (ICP) algorithm for aligning datasets of unstructured points (Reddy et al. 2004; Zhong et al. 2005; Dong et al. 2006; Zhong et al. 2006). The registration accuracy hinges on both the number and diversity of points informing the ICP algorithm. Using CartoMerge, Dong et al. (2006) reported an average registration error of around 3 mm, while the average time required for mapping and registration was about 40 minutes.

A major problem with EPS is that patient motion can cause the reference electrode to move, resulting in measured catheter tip positions falling outside the atrial point cloud model or registered mesh. Should this happen, an entirely new point cloud model must be rebuilt. This can be particularly frustrating as data collection often takes up to an hour, and is complex enough to warrant expert supervision. Less severe but equally problematic, EPS measurements may drift over time due to a variety of factors including the presence of ferrous metals in the operating room, or changes in the patient's atrial impedance profile during the procedure. Recent studies have shown the potential for using ICE ultrasound in combination with EPS to speed up the process of reacquiring data for anatomic mesh registration (Zhong et al. 2006; Wein et al. 2007). The issue of how to deal with reference electrode movement or drifting remains unresolved, although acquiring data can be substantially less time consuming using this approach (Zhong et al. 2006).

4. Methods

Our goal is to automatically estimate the pose of the ICE catheter inside the left atrium as accurately as commercial EPS mapping systems, but in a fraction of the time (around 3 mm and about 40 minutes, as reported by Dong et al. (2006)). To estimate the ICE catheter's pose, we use a recursive Bayesian state estimation (RBSE) framework commonly used in mobile robot localization (Thrun et al. 2005). Unlike many medical image registration algorithms, the RBSE framework expresses its estimate as a probability distribution over a range of possible solutions. With built-in facilities for accommodating measurement uncertainty and deviations from expected conditions, RBSE is an ideal candidate for intraoperative ultrasound registration.

The RBSE framework tracks the evolution of three random variables over time:

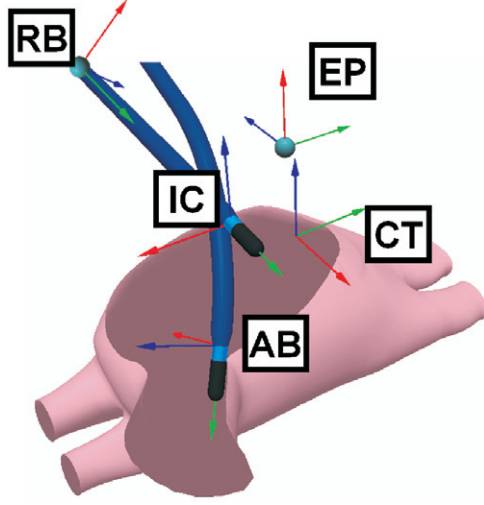


Fig. 5. Reference frames. Coordinate axes are defined for the robotic catheter base (RB), EPS (EP), ICE catheter tip (IC), ablation catheter tip (AB), and surface mesh (CT) reference frames.

1. the *state* variable $\mathbf{x}$, which describes the parameters we wish to estimate;
2. the *control* variable $\mathbf{u}$ (and associated state transition model f_{tran}), which describes inputs that alter the state; and
3. the *measurement* variable $\mathbf{z}$ (and associated measurement model f_{meas}), which describes measurements for observing the state.

In the following sections, we detail our choice of variables and the specifics of our RBSE implementation.

4.1. State Parameters

We define state variable $\mathbf{x}$ to be the *pose* (position and orientation) of the ICE catheter with respect to the left atrium. We use reference frame formalisms to codify our knowledge of state. In Figure 5 we see reference frames assigned to the robotic guide catheter base (RB), electromagnetic position sensor base (EP), ICE catheter tip (IC), ablation catheter tip (AB), and CT/MR-based surface mesh (CT). Our state estimate is equivalent to the relationship between reference frames CT and IC.

As described in Section 2, our algorithm compares ICE images of the anatomy to virtual ICE images of the surface mesh, thereby implicitly assuming that the mesh represents the left atrium. As a consequence of this assumption, localizing the ICE catheter tip with respect to the surface mesh equates to localizing the ICE catheter tip in the left atrium.

To relate reference frames A and B , we define affine transformations $T \in \mathbb{R}^{4 \times 4}$ with the following notation:

$${}^A T^B = \begin{bmatrix} {}^A R^B & {}^A O^B \\ 0 & 1 \end{bmatrix}, \quad (1)$$

where ${}^A R^B \in \mathbb{R}^{3 \times 3}$ and ${}^A O^B \in \mathbb{R}^{3 \times 1}$ are the rotation and origin offsets between the coordinate axes of A and B . We represent ${}^A T^B$ by its pose vector ${}^A P^B \in \mathbb{R}^{6 \times 1}$:

$${}^A P^B = [x, y, z, \alpha, \beta, \gamma]^T, \quad (2)$$

where ${}^A O^B = [x, y, z]^T$ and α, β , and γ are the xyz Euler angles used to compute ${}^A R^B$. Finally, we define function $h_{P \rightarrow T} \in \mathbb{R}^{6 \times 1 \rightarrow 4 \times 4}$ to convert ${}^A P^B$ into ${}^A T^B$:

$$h_{P \rightarrow T} = \begin{bmatrix} & & & x \\ & R_z(\gamma) & R_y(\beta) & R_x(\alpha) & y \\ & & & z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

with corresponding inverse $h_{T \rightarrow P} \in \mathbb{R}^{4 \times 4 \rightarrow 6 \times 1}$. Using this notation, our state estimate at time k is expressed simply as

$$\mathbf{x}_k = {}^{CT} P^{IC_k}. \quad (4)$$

4.2. Control Parameters and State Transition Model

We define the control variable $\mathbf{u}$ to be the change in pose of the ICE catheter tip between times $k-1$ and k . This is mathematically described using the pose transformation between the IC reference frame at time $k-1$ (IC_{k-1}) and at time k (IC_k):

$$\mathbf{u}_k = {}^{IC_{k-1}} P^{IC_k}. \quad (5)$$

The associated state transition model f_{tran} is defined as

$$\begin{aligned} \mathbf{x}_k &= f_{\text{tran}}(\mathbf{x}_{k-1}, \mathbf{u}_k) \\ &= h_{T \rightarrow P}(h_{P \rightarrow T}(\mathbf{x}_{k-1})h_{P \rightarrow T}(\mathbf{u}_k)). \end{aligned} \quad (6)$$

To compute $\mathbf{u}_k$ we employ one of two strategies. The first is to use an EPS sensor attached to the IC reference frame and measure its movement with respect to the EP reference frame. Specifically, we use a six-degree-of-freedom (DOF) magnetic position/orientation sensor (Ascension Tech, Burlington, VT) whose position resolution is 0.5 mm along each axis, and orientation resolution is 0.1° about each axis. As demonstrated in Figure 6, we incorporate the pose uncertainty related to the sensor's limited resolution by modeling $\mathbf{u}$ as a Gaussian random variable:

$$\mathbf{u}_k = \mathcal{N}(\mu_{u_k}, \Sigma_{u_k}), \quad (7)$$

where the expected change in pose is

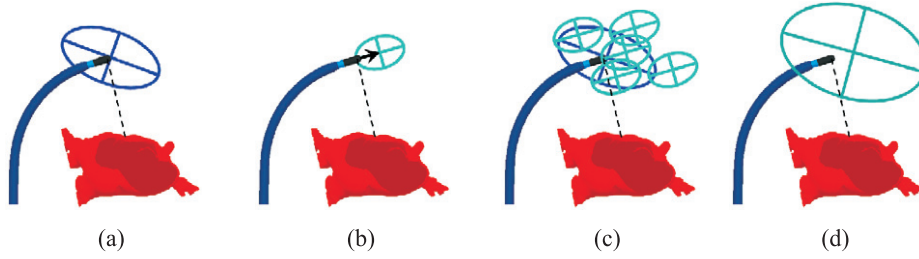


Fig. 6. Incorporating kinematic model uncertainty. In (a) we represent the uncertainty in our belief of catheter tip pose with respect to the left atrium as an ellipsoid. In (b) we see the uncertainty in our kinematic model when moving the catheter tip in the direction indicated by the arrow. In (c) we combine our pose and kinematic model uncertainties to yield the ellipsoid in (d), which represents our pose uncertainty after moving the catheter, but before acquiring measurements to improve our estimate.

$$\mu_{u_k} = h_{T \rightarrow P} \left(\left({}^{IC_{k-1}}T^{EP} \right) \left({}^{EP}T^{IC_k} \right) \right) \quad (8)$$

and the transition uncertainty is

$$\Sigma_{u_k} = \text{diag} (.5 \text{ mm}, .5 \text{ mm}, .5 \text{ mm}, .1^\circ, .1^\circ, .1^\circ) \quad (9)$$

The second strategy for computing $\mathbf{u}_k$ is to employ the robotic guide catheter's kinematics in determining the movement of IC with respect to the RB reference frame. As before, we model $\mathbf{u}$ as a Gaussian random variable, and use the robotic guide's joint commands to compute μ_{u_k} and Σ_{u_k} .

4.2.1. Kinematics-based Expected Change in Pose

The robotic guide catheter (Hansen Medical, Inc., Mountain View, CA) is a three-DOF system capable of full-body rotation and planar tip deflection (Figure 7(a)). The ICE catheter is inserted through a lumen running up the center of the guide, and its tip emerges at the lumen's distal end. This enables the ICE catheter to rotate independently of the guide and image a swept volume from a fixed position, as illustrated in Figure 7(b).

The kinematic model that defines ${}^{RB}T^{IC}$ is shown in Figure 8. Briefly, the robotic guide's rotation angle σ , also known as the *roll* angle, is applied axially at its base. While *rolling*, the entire guide moves as a single rigid body. Tip deflection is accomplished by deforming the guide catheter using compressive forces acting along the guide's body. As described in Camarillo et al. (2008), these compressive forces cause uniform bending along two segments, resulting in bends that occur along arcs of constant curvature. The arc angles ϕ and ψ are referred to as the *proximal* and *distal bend* angles. In combination, the two bend angles control the guide's off-axis tip position and heading. The ICE catheter tip tracks the guide's tip position, but not its axial rotation; this is because the ICE catheter's rotation angle θ is controlled independently. Therefore, in our kinematic model, we subtract out roll angle σ when computing the ICE catheter tip's final orientation.

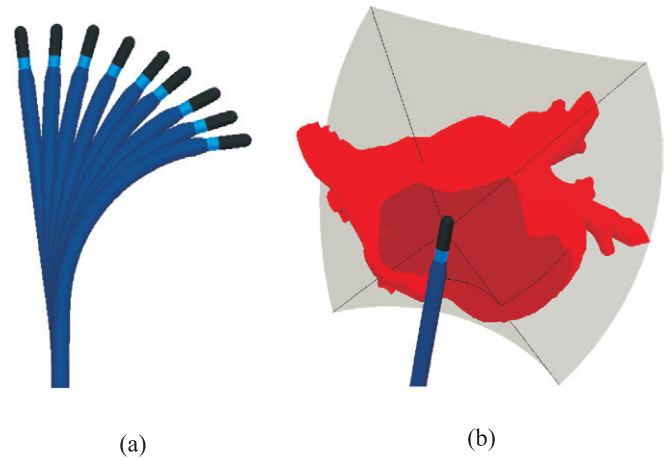


Fig. 7. Catheter tip deflection and rotation. In (a) we see a range of tip deflection angles that can be applied to the robotic guide for steering the ICE catheter tip to a number of different positions and orientations. In (b) we demonstrate how the ICE catheter can rotate independently within the robotic guide's lumen to image a swept volume.

The left part of Figure 8 introduces intermediate frames J_1 , J_2 , and J_3 between the actuated joints of our system. On the right and bottom of Figure 8, we define coordinate origin and rotation offsets for computing the transformation matrices between these intermediate frames. When bending angles ϕ and ψ are small, ${}^{J_1}O^{J_2}$ and ${}^{J_2}O^{J_3}$ are linearized to avoid singularities.

Finally, the transformations linking the intermediate frames are multiplied to obtain ${}^{RB}T^{IC}$:

$${}^{RB}T^{IC} = ({}^R T^{J_1}) ({}^{J_1} T^{J_2}) ({}^{J_2} T^{J_3}) ({}^{J_3} T^{IC}) \quad (10)$$

and the expected change in pose at time k is

$$\mu_{u_k} = h_{T \rightarrow P} \left(\left({}^{IC_{k-1}}T^{RB} \right) \left({}^{RB}T^{IC_k} \right) \right). \quad (11)$$

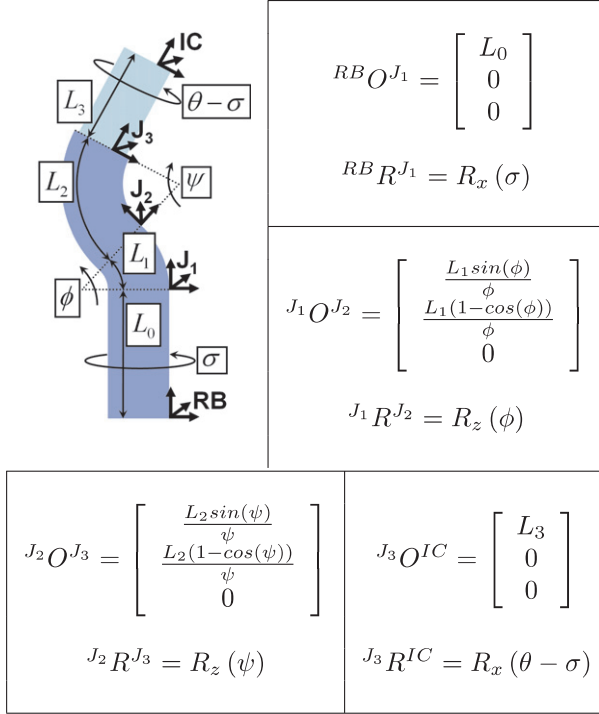
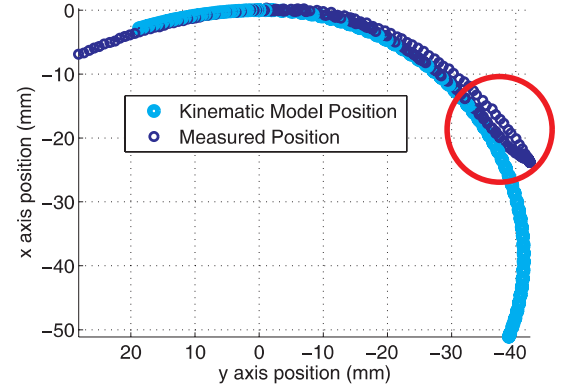


Fig. 8. The kinematic model describes the catheter tip reference frame IC with respect to the robotic guide catheter base RB . We introduce intermediate frames J_1 , J_2 , and J_3 and decompose the model into four separate transformations, defined by the coordinate axes origin and rotation offsets above. Parameters L_0 , L_1 , L_2 , and L_3 are the respective lengths of the individual robotic guide segments.

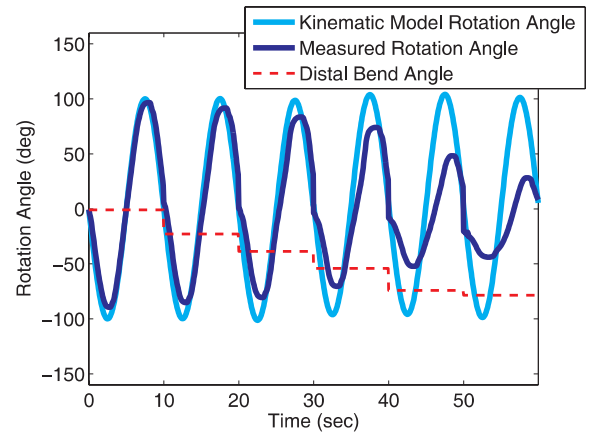
4.2.2. Kinematics-based Transition Uncertainty

The kinematic model described above works reasonably well over most of the guide catheter's workspace, but can render poor ${}^{RB}T^{IC}$ estimates at the workspace boundaries. For example, the arc of constant curvature assumptions used to characterize planar bending relies on the guide exhibiting elastic behavior. In reality, as the guide tip is deflected and then straightened out, its body undergoes plastic deformation causing hysteresis and creep. In Figure 9(a) we see the model's position estimate and measured trajectory over a range of tip deflection angles (as shown in Figure 7(a)). As the deflection angle increases, so does the divergence between expected and actual behavior. Furthermore, we see that during repeated bending, the measured trajectory varies slightly each repetition.

To complicate matters further, the kinematic model's accuracy is configuration dependent. For example, the rotation of the ICE catheter inside the guide's lumen is subject to friction whose effects are dependent on the amount of guide tip deflection. Furthermore, as the guide bends it also compresses, causing the ICE catheter's tip to protrude further out of the guide's lumen.



(a)



(b)

Fig. 9. Kinematic model errors. In (a) we see the estimated and actual tip position trajectory as the guide catheter is deflected and straightened over several cycles. Trajectory error increases along with bending angle. The measured trajectory also varies slightly each cycle, as highlighted in the circled area. In (b) we see the effect of guide distal bend (ψ) on ICE catheter rotation (θ) tracking.

Figure 9(b) shows the inaccuracies in commanded and measured ICE catheter rotation angle as a function of guide distal bend. As the magnitude of the distal bend angle increases, so does the divergence between the expected and actual behavior.

A more accurate kinematic model than what is presented in the previous section could be derived to account for these numerous inaccuracies, but doing so would require an exhaustive characterization of the entire catheter workspace. Even if this was tractable, the model would still be subject to additional errors incurred once the catheter is inserted into the anatomy. Instead of deriving a more complex kinematic model, we incorporate deviations from the model into our transition uncertainty matrix Σ_{u_k} .

To quantify Σ_{u_k} , we exhaustively drove the catheter over its entire workspace while taking measurements of the ICE catheter tip pose using a magnetic position sensor. For simplicity, we assumed that only one joint angle could be driven at a time. For each non-zero command issued to joint j , $command_{j_i} \in \mathbb{R}^1$, we computed the error, $error_{j_i} \in \mathbb{R}^{6 \times 1}$, between the expected and actual change in catheter tip pose. Then, for each joint j , we derive a normalized command error covariance $\Sigma_j \in \mathbb{R}^{6 \times 6}$ as follows:

$$\Sigma_j = \frac{\sum_{i=1}^N \left(\frac{error_{j_i}}{|command_{j_i}|} \right) \left(\frac{error_{j_i}}{|command_{j_i}|} \right)^T}{N}. \quad (12)$$

When running our RBSE algorithm, we compute the transition uncertainty Σ_{u_k} as a scaled sum of the individual joint command error covariances:

$$\Sigma_{u_k} = \Sigma_{\text{drift}} + \sum_{j=1}^4 \left((\Sigma_j) (|command_{k,j}|^2) \right), \quad (13)$$

where

$$command_k = [\delta\sigma, \delta\phi, \delta\psi, \delta\theta]^T \quad (14)$$

and $command_{k,j}$ is the specific command issued to joint j . Here $\Sigma_{\text{drift}} \in \mathbb{R}^{6 \times 6}$ is a noise matrix that incorporates additional uncertainty in our state transition model unrelated to kinematic model errors. Sources for additional uncertainty might include anatomy shifts caused by breathing or sudden patient movement. For the purpose of this study we assume that anatomy shifts are small and independent between pose variables, although we could be more precise in constructing Σ_{drift} to incorporate knowledge we have regarding the expected nature of anatomy shifts.

4.3. Measurement Parameters and Model

The acquired $n \times m$ ICE image serves as our measurement variable $\mathbf{z}_k \in \mathbb{R}^{n \times m}$. Our measurement estimate, $\mathbf{y}_k \in \mathbb{R}^{n \times m}$, is generated using measurement model f_{meas} which computes a virtual ICE image based on state estimate $\mathbf{x}_k$:

$$\mathbf{y}_k = f_{\text{meas}}(\mathbf{x}_k). \quad (15)$$

In the ensuing sections we describe our method for generating virtual ICE images, as well as the calibration procedure needed to ensure that the virtual ICE images approximate reality.

4.3.1. Surface Mesh and Phantom Generation

As noted previously, our algorithm assumes the surface mesh used for generating virtual ICE images equivalently represents the left atrium. For this reason, it is important that the geometry

of the surface mesh approximates that of the anatomy imaged under ultrasound.

We generate the surface mesh by segmenting the left atrium from a cardiac CT scan using Verizmo (St Jude Medical, Minneapolis, MN), a software platform for CT segmentation widely employed in EP medicine. We can reasonably assume that a significant percentage of cardiac CT scans acquired in advance of a catheter ablation procedure are segmented using Verizmo. We therefore consider Verizmo to be an accepted means of segmentation.

To test our algorithm *in vitro*, we created an ultrasound phantom using the surface mesh as a geometric reference. The phantom is a cavity-mold cast in 0.1% silicone dioxide, a composite material that exhibits tissue-like reflectivity when imaged under ultrasound. The surface of the cavity-mold was bead-blasted to generate texture for improved reflections.

4.3.2. Occupancy Grid Generation

Our algorithm for generating virtual ICE images is based on sampling cross sections of the CT surface mesh. This task is simplified by representing the surface mesh as an occupancy grid (Figure 10(a)). To generate the occupancy grid, we first shift the mesh vertex coordinates so that the CT reference frame origin lies inside the mesh. Then we define a three-dimensional grid with resolution and dimensions chosen to fully encompass the extent of the mesh. To determine whether grid coordinate

$$C_{G_i} = [x_{c_{g_i}}, y_{c_{g_i}}, z_{c_{g_i}}]^T$$

lies inside or outside the mesh, we project an imaginary ray outwards from C_{G_i} and count the number of times the ray intersects a mesh face. An odd number of intersections implies that C_{G_i} lies inside the mesh, and an even number implies that C_{G_i} lies outside the mesh. Coordinates inside the mesh are assigned a value of zero, and coordinates outside the mesh are assigned a value of one. In this way our occupancy grid identically represents the spatial distribution of silicone dioxide in our phantom cavity mold.

4.3.3. Virtual ICE Generation

Virtual ICE images are generated by sampling planar cross sections of the CT occupancy grid map (Figure 10(a)). First, an $n \times m$ virtual image plane is defined with dimensions and scaling equivalent to that of the frame-grabbed ICE images. Each coordinate

$$C_{V_i} = [x_{c_{v_i}}, y_{c_{v_i}}]^T$$

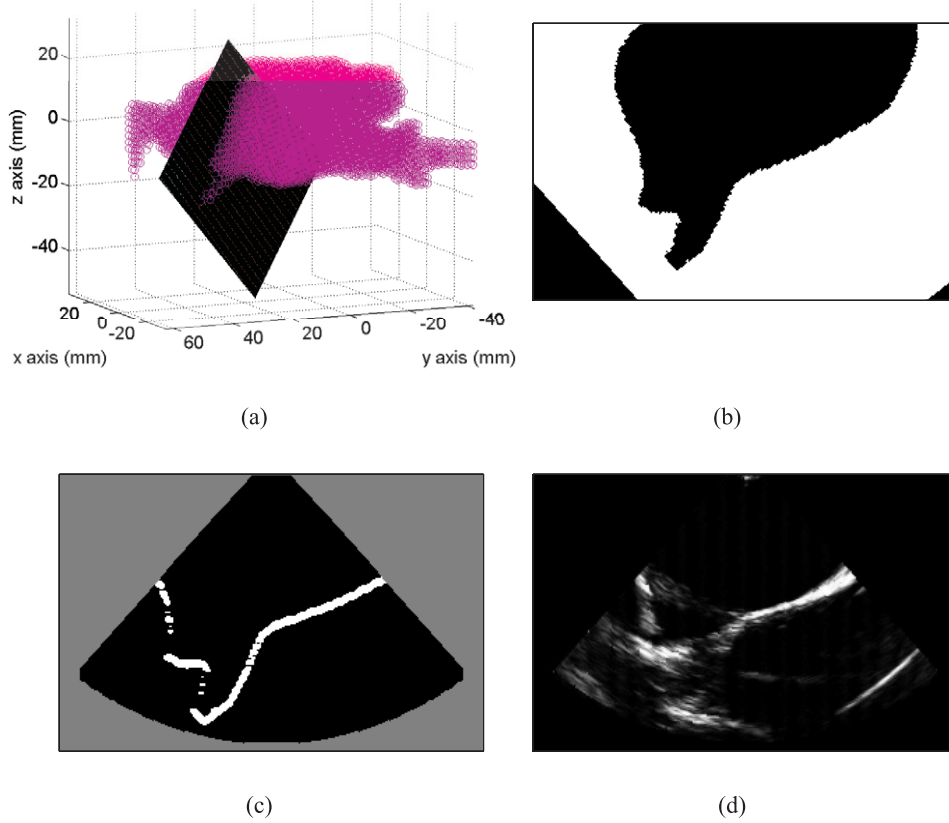


Fig. 10. Virtual ICE generation from an occupancy grid. In (a) we see an occupancy grid generated from a CT surface mesh of the left atrium. The plane that intersects with the occupancy grid is used to generate a cross-sectional image of the anatomy (b). The cross-sectional image is filtered using an edge-detection kernel to generate a virtual ultrasound image (c) that roughly approximates an actual ultrasound image taken of the left atrium phantom (d).

of the virtual image is transformed into a occupancy grid coordinate C_{G_i} using the equation

$$\begin{bmatrix} x_{c_{g_i}} \\ y_{c_{g_i}} \\ z_{c_{g_i}} \\ 1 \end{bmatrix} = \begin{pmatrix} {}^{CT}T^{IC} \end{pmatrix} \begin{pmatrix} {}^{IC}T^{US} \end{pmatrix} \begin{bmatrix} x_{c_{v_i}} \\ y_{c_{v_i}} \\ 0 \\ 1 \end{bmatrix}. \quad (16)$$

We use nearest-neighbor interpolation to assign the pixel value (zero or one) to C_{v_i} based on C_{G_i} , resulting in an image such as that shown in Figure 10(b).

The planar cross section is filtered using a simplified model for ultrasound reflections. Briefly, we assume that an ultrasound wave will generate a reflection at the interface of two different mediums provided that the medium in the wave's forward path has a reflective surface. Reflection strength is determined by the perpendicularity between the reflective surface and the wave path. To simulate this behavior, we run an edge-detection filter on the planar cross section from top to bottom.

The filter yields positive values at the transition between free space and tissue, negative values at the intersection of tissue and free space, and zero otherwise. Finally we threshold the filtered image, yielding a virtual ICE image that closely approximates an ICE image obtained from imaging the phantom (Figures 10(c) and 10(d)).

A more accurate filtering technique would be to detect edges along the paths of the ultrasound scan lines, which travel radially outwards from the ICE transducer. However, because we need to generate several thousand virtual ICE images at each time step, it is significantly faster to use a simple linear filter. This approach works for *in vitro* testing, but a more sophisticated approach, such as that described by Wein et al. (2007), may be required for *in vivo* testing.

4.3.4. Ultrasound Acquisition and Calibration

ICE images of the phantom were captured using an AcuNav ICE catheter attached to a Sequoia Ultrasound System

(Siemens Medical, Portland, OR). The images were frame-grabbed onto a PC using a Foresight I-Color frame-grabber (Foresight, Boston, MA) at a rate of 15 Hz.

We performed two calibration procedures to determine the parameters relating image pixel coordinates to real-world (mm) coordinates. To assist in this process, we define two additional reference frames: the image reference frame (IM) and the ultrasound reference frame (US). The US reference frame is distinct from the IC reference frame in that the IC reference frame is where the EPS sensor is mounted to the ICE catheter tip, and the US reference frame is where the actual ultrasound transducer is located inside the ICE catheter.

The first calibration procedure solves for the *intrinsic parameters* defining ${}^{US}T^{IM}$, the transformation that takes image pixel coordinates and converts them into real-world coordinates in the US reference frame. We fix the IM reference frame origin to the top left corner of the frame-grabbed ICE image, and the US reference frame origin to the point where the left and right edges of the ultrasound sector intersect. We denote the image pixel coordinate of the US reference frame origin as $C_0 = [x_{c_0}, y_{c_0}]^T$.

To determine pixel scaling parameters α_x and α_y (which are measures of pixels per millimeter in the $\hat{x}_{IM}$ and $\hat{y}_{IM}$ directions) we image a phantom with seven wires ($W_1, \dots, W_7$) under ultrasound, yielding an image with seven circles whose centroids are located at pixels $C_{w_1}, \dots, C_{w_7}$. The real-world coordinates where the W_i intersect the ultrasound plane can be related to their corresponding image coordinates C_{w_i} using the equation:

$$\begin{bmatrix} x_{c_{w_i}} \\ y_{c_{w_i}} \\ 0 \\ 1 \end{bmatrix} = A \cdot T \cdot \begin{bmatrix} x_{w_i} \\ y_{w_i} \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} x_{c_0} \\ y_{c_0} \\ 0 \\ 1 \end{bmatrix}, \quad (17)$$

where

$$A = \begin{bmatrix} \alpha_x & 0 & 0 & 0 \\ 0 & \alpha_y & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (18)$$

and T is the transformation matrix that defines the rotation and translation parameters for aligning the phantom and image reference frames:

$$T = \begin{bmatrix} R_{11} & R_{12} & 0 & O_x \\ R_{21} & R_{22} & 0 & O_y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (19)$$

We employ least-squares estimation to solve for α_x and α_y using the seven (C_{w_i}, W_i) using the equation:

$$\begin{bmatrix} \alpha_x R_{11} \\ \alpha_x R_{12} \\ \alpha_x O_x \\ \alpha_y R_{21} \\ \alpha_y R_{22} \\ \alpha_y O_y \end{bmatrix} = M_{LS}^\dagger \begin{bmatrix} x_{c_{w_1}} - x_{c_0} \\ y_{c_{w_1}} - y_{c_0} \\ \vdots \\ x_{c_{w_7}} - x_{c_0} \\ y_{c_{w_7}} - y_{c_0} \end{bmatrix}, \quad (20)$$

where

$$M_{LS} = \begin{bmatrix} x_{w_1} & y_{w_1} & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x_{w_1} & y_{w_1} & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{w_7} & y_{w_7} & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x_{w_7} & y_{w_7} & 1 \end{bmatrix} \quad (21)$$

$$\alpha_x = \sqrt{(\alpha_x R_{11})^2 + (\alpha_x R_{12})^2}, \quad (22)$$

$$\alpha_y = \sqrt{(\alpha_y R_{21})^2 + (\alpha_y R_{22})^2}. \quad (23)$$

Finally, we compute ${}^{US}T^{IM}$ as

$${}^{US}T^{IM} = \begin{bmatrix} \frac{1}{\alpha_x} & 0 & 0 & -\frac{1}{\alpha_x} x_{c_0} \\ 0 & \frac{1}{\alpha_y} & 0 & -\frac{1}{\alpha_y} y_{c_0} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (24)$$

We measure the accuracy of our intrinsic parameter calibration by computing the error between the expected pixel coordinates for the W_i and their actual coordinates. Specifically, for each

W_i we compute the normalized scan-line distance (r) and angle (ϑ) errors as

$$r_{\text{err}} = \frac{|r_{W_i}^{\text{exp}} - r_{W_i}^{\text{act}}|}{r_{W_i}^{\text{exp}}}, \quad (25)$$

$$\vartheta_{\text{err}} = \frac{|\theta_{W_i}^{\text{exp}} - \theta_{W_i}^{\text{act}}|}{\theta_{W_i}^{\text{exp}}}. \quad (26)$$

The average scan-line distance and angle errors are $r_{\text{err}} = 0.009$ mm/mm and $\vartheta_{\text{err}} = 0.001^\circ/^\circ$. For an ICE image acquired with 100 mm of depth, the real-world coordinate estimate error for a corner pixel is approximately 0.94 mm.

The second calibration procedure solves for the *extrinsic parameters* defining ${}^{IC}T^{US}$. To determine these parameters, we use a single-target imaging technique described by Detmer et al. (1994).

4.4. RBSE Implementation

At time step k , the RBSE state estimate is defined as the *posterior* distribution of state $\mathbf{x}_k$ conditioned on all past controls $\mathbf{u}_{1:k}$ and measurements $\mathbf{z}_{1:k}$ (Thrun 2000):

$$f_{\text{est}}(\mathbf{x}_k) = p(\mathbf{x}_k | \mathbf{u}_{1:k}, \mathbf{z}_{1:k}). \quad (27)$$

Evaluating f_{est} is made tractable by assuming that state variable $\mathbf{x}$ is a Markov process. Under this assumption we can define f_{est} recursively as

$$f_{\text{est}}(\mathbf{x}_k) = \left(\eta \cdot p(\mathbf{z}_k | \mathbf{x}_k) \times \int p(\mathbf{x}_k | \mathbf{x}_{k-1}, \mathbf{u}_k) f_{\text{est}}(\mathbf{x}_{k-1}) \delta \mathbf{x}_{k-1} \right) \quad (28)$$

with normalization constant η (Thrun 2000). In the event that $f_{\text{est}}(\mathbf{x}_k)$ is Gaussian and both f_{meas} and f_{tran} are linearizable, f_{est} can be evaluated using either the Kalman or extended Kalman filter. However, because our measurement and state transition models are highly non-linear, and because $f_{\text{est}}(\mathbf{x}_k)$ is decidedly non-Gaussian (Figure 11), we employ the unscented particle filter (UPF). The UPF models the posterior likelihood of $\mathbf{x}$ using particle set $S = \{s_1, \dots, s_N\}$, where each particle s_i encapsulates an instantiation of $\mathbf{x}$, $s_i^x \in \mathbb{R}^{6 \times 1}$ (van de Merwe et al. 2001). The particles are assumed to have a locally Gaussian distribution $s_i^x \in \mathcal{N}(\mu_{s_i^x}, \Sigma_{s_i^x})$, allowing each particle to cover a large region of the configuration space. This way we can adequately cover the distribution of $f_{\text{est}}(\mathbf{x}_k)$ without having to instantiate a number of particles exponential in the dimensions of the configuration space.

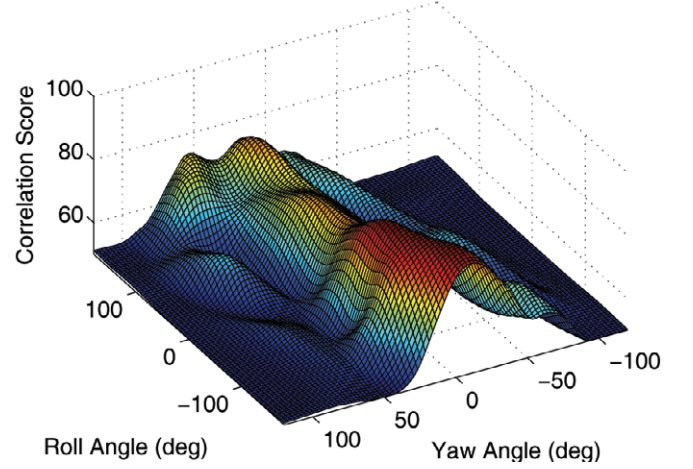


Fig. 11. State posterior distribution. Assuming that four of the six parameters defining $\mathbf{x}_k$ are known (x , y , z , and β), we compute the state posterior distribution over the remaining two parameters (α and γ). As the distribution is non-Gaussian, we cannot employ Gaussian state estimation techniques.

Each particle in our UPF implementation is initially assigned a random pose distribution taken from a uniform distribution over all possible poses, the range of which is assumed to be slightly larger than the size of an adult left atrium:

$$\mathbf{x}_{\text{range}} = \begin{bmatrix} x_{\text{range}} \\ y_{\text{range}} \\ z_{\text{range}} \\ \alpha_{\text{range}} \\ \beta_{\text{range}} \\ \gamma_{\text{range}} \end{bmatrix} = \begin{bmatrix} \pm 50 \text{ mm} \\ \pm 50 \text{ mm} \\ \pm 50 \text{ mm} \\ \pm 180^\circ \\ \pm 90^\circ \\ \pm 120^\circ \end{bmatrix}. \quad (29)$$

The initial distribution for each particle is then

$$s_{i0}^x \in \mathcal{N}(\mu_{s_{i0}^x}, \Sigma_{s_{i0}^x}), \quad (30)$$

where

$$\mu_{s_{i0}^x} = \text{rand}_U(\mathbf{x}_{\text{range}}). \quad (31)$$

and

$$\Sigma_{s_{i0}^x} = \text{diag} \left(\max(\mathbf{x}_{\text{range}}) - \min(\mathbf{x}_{\text{range}}) \right)^2. \quad (32)$$

At time k , each particle evolves using the unscented Kalman filter algorithm described by Julier and Uhlmann (1997). A key modification is that the measurement residual between the estimated virtual ICE image and the acquired ICE image is computed as a scalar correlation score. The specifics of the image correlation algorithm are described in the following section.

After all particles have evolved, S is resampled taking into consideration each particle's image correlation score, s_i^c , using a variant of the stochastic universal sampling technique described by Baker (1987). Particles with low scores are discarded and eventually only a few particles, whose estimates are close to the actual state, remain in S .

4.4.1. Measurement Residual Computation

As our measurement variable is an image, we use an image correlation metric to evaluate our measurement residual. Specifically, we use a normalized mutual information (NMI) correlation metric, as described in Pluim et al. (2003). NMI allows us to measure the similarity between images not just on the basis of how well their pixels match up, but on the basis of how much "content" is in the image as well. Furthermore, NMI operates only on image pixels, which is desirable because of the difficulty in reliably segmenting cardiac ultrasound images.

Mutual information is rooted in the idea of measuring the information content of a signal. Here the signal is the image, and the information is comprised of the image pixel values. The amount of information we can infer from the image depends on how we expect the image to look. For example, if we compare two images that both have a large number of black pixels, a few overlapping white pixels conveys a large amount of information. Conversely, if we compare two images that have similar numbers of black and white pixels, a few overlapping white pixels does not convey very much information at all.

As our virtual ICE images are binary, we threshold the acquired ICE images to binary so the range of pixel values are identical between the two image types. Doing so also speeds up the NMI correlation computation by allowing us to perform comparisons using bitwise operations. We also decimate the resolution of the acquired ICE image when there are a large number of particles to consider, and gradually return to the native resolution as particles are discarded.

4.4.2. Particle Set Resampling

In the standard particle filter implementation, multiple copies of s_i may be resampled into S if its measurement residual indicates high likelihood. However, these copies will diverge in the next time step as variance in the state transition model is introduced during particle evolution. As UPF particles represent a distribution of configurations rather than a single, unique configuration, retaining multiple copies of the same particle is redundant; even after the copies diverge in the next time step, their distributions will likely overlap. Milstein et al. (2002) and van der Merwe and Wan (2003) propose representing particle set S as a collection of Gaussian clusters. The difficulty in

this approach is knowing how many clusters to create. Rather than using this approach, we assign weights s_i^w to each particle based on the number of times a copy of s_i was resampled, N_{s_i} . The weights are computed as

$$s_i^w = \frac{N_{s_i}}{N_{\max}} \quad (33)$$

where $N_{\max}$ is the total number of particles resampled. Therefore, the number of unique particles, N , will always be less than $N_{\max}$, and $\sum_{i=1}^N s_i^w = 1$. After assigning these weights, we discard copies and retain only unique particles. At the next time step, we multiply each particle's current weight by its image correlation score and resample on the basis of this product.

As our state posterior distribution is both multi-modal and sparse in between peaks, there is a reasonable risk for particle deprivation. This occurs when all particles start out in a region of negligible likelihood or near a local maxima; these particles will either be discarded or converge upon an incorrect solution. In order to recover from such a scenario, we employ the augmented particle filter strategy described by Gutmann and Fox (2002). Briefly, at time k we calculate a confidence estimate:

$$f_{\text{confidence}}(S_k) = \sum_{i=1}^N s_{i_k}^w s_{i_k}^c \quad (34)$$

and maintain slow-moving and fast-moving filtered averages (κ_{slow} and κ_{fast}) of our confidence estimate over time. When κ_{fast} drops below κ_{slow} , we introduce new particles into the particle set with probability:

$$p_{\text{new}} = 1 - \frac{\kappa_{\text{fast}}}{\kappa_{\text{slow}}}. \quad (35)$$

We first resample the existing particle set into $S_{\text{old}} = \{s_{1_{\text{old}}}, \dots, s_{N_{\text{old}}}\}$. Then we create a new set of equally weighted particles $S_{\text{new}} = \{s_{1_{\text{new}}}, \dots, s_{N_{\text{new}}}\}$, where

$$\frac{N_{\text{new}}}{N_{\text{old}} + N_{\text{new}}} \propto 1 - \frac{\kappa_{\text{fast}}}{\kappa_{\text{slow}}}. \quad (36)$$

Finally, we merge the two sets and reassign weights as follows:

$$\overline{s_{i_{\text{old}}}^w} = \frac{N_{\text{old}}}{N_{\text{old}} + N_{\text{new}}} s_{i_{\text{old}}}^w, \quad (37)$$

$$\overline{s_{i_{\text{new}}}^w} = \frac{N_{\text{new}}}{N_{\text{old}} + N_{\text{new}}} s_{i_{\text{new}}}^w, \quad (38)$$

which ensures that

$$\sum_{i=1}^{N_{\text{old}}} \overline{s_{i_{\text{old}}}^w} + \sum_{i=1}^{N_{\text{new}}} \overline{s_{i_{\text{new}}}^w} = 1. \quad (39)$$

The pose distributions of the new particles are initialized to $s_{i_{\text{new}}}^x \in \mathcal{N}(\mu_{s_{\text{new}}}^x, \Sigma_{s_{\text{new}}}^x)$ where

$$\mu_{s_{\text{new}}}^x = f_{\text{est}}(\mathbf{x}_k) + \text{rand}_N \left((1 - \kappa_{\text{fast}}) \Sigma_{s_{\text{init}}}^x \right) \quad (40)$$

and

$$\Sigma_{s_{\text{new}}}^x = (1 - \kappa_{\text{fast}}) \Sigma_{s_{\text{init}}}^x. \quad (41)$$

Simply put, the pose distributions of the new particles are scattered about the current state estimate in accordance to the current fast-moving confidence average. The less confident the average, the larger the distribution.

4.4.3. State Estimate Expected Value and Covariance

Finally, at time step k we compute the expected value of our state as a weighted sum of the individual particle estimates μ_{s_i} :

$$\mathbf{E}[\mathbf{x}_k] = \sum_{i=1}^N s_i^w \mu_{s_i} \quad (42)$$

and compute the covariance of our state estimate as

$$\text{Cov}[\mathbf{x}_k] = \sum_{i=1}^N s_i^w \Sigma_{s_i}. \quad (43)$$

When evaluating our localization algorithm's performance in Section 5, we use the expected value to compute our estimate error at each time step.

5. Results

5.1. Experimental Setup

We verified our UPF localization algorithm's performance by driving an ICE catheter into a left atrium phantom and estimating the catheter's pose on the basis of acquired ICE images.

The ICE catheter was driven using the robotic catheter control system shown in Figure 12(a), and its handle was attached to a motorized rotation system allowing it to rotate independently of the robotic guide. This enabled imaging of swept volumes from a fixed ICE probe position.

A six-DOF electromagnetic position/orientation sensor was attached to the ICE catheter's tip to measure ${}^{EP}T_{\text{meas}}^{IC}$ (Figure 12(b)), and a second sensor was attached to the phantom to measure ${}^{EP}T_{\text{meas}}^{CT}$. The measurement of state variable $\mathbf{x} = {}^{CT}P^{IC}$ is computed as

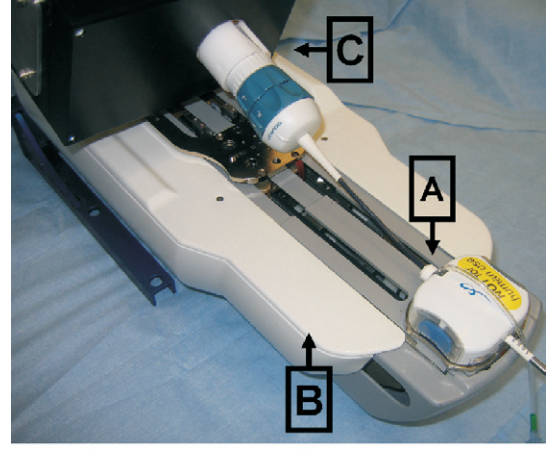
$${}^{CT}P_{\text{meas}}^{IC} = h_{T \rightarrow P} \left(({}^{CT}T_{\text{meas}}^{EP}) ({}^{EP}T_{\text{meas}}^{IC}) \right). \quad (44)$$

5.1.1. EPS to Mesh Registration

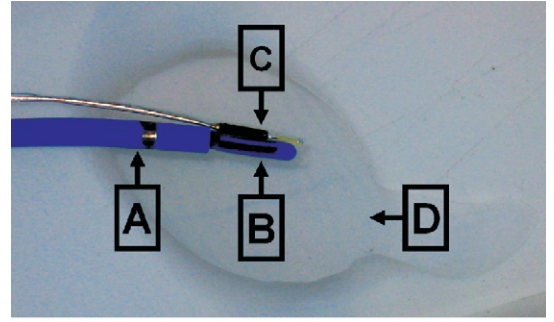
Using our UPF estimate of ${}^{CT}P^{IC_k}$, we recover an estimate of ${}^{CT}T^{EP_k}$ in one of two ways. In general, we assume that the EPS reference frame may drift over time without our knowledge. In this case we can directly compute ${}^{CT}T^{EP_k}$ as

$${}^{CT}T^{EP_k} = h_{P \rightarrow T} ({}^{CT}P^{IC_k}) {}^{IC}T_{\text{meas}}^{EP}. \quad (45)$$

However, if we assume that ${}^{CT}T^{EP_k}$ drifts slowly, we can compute a least-squares estimate of ${}^{CT}T^{EP_k}$ by considering the last



(a)



(b)

Fig. 12. Robotic catheter control system. In (a) we see the ICE catheter inserted through the robotic guide's lumen (A), the controller used to drive the guide (B), and a mechanism for rotating the ICE catheter axially (C). In (b) we see the robotic guide (A) with ICE catheter tip emerging from its lumen (B). A sensor (C) is mounted to the ICE catheter's tip for validation measurements. The ICE catheter images a left atrium phantom (D) serving as an anatomical proxy.

n UPF estimates simultaneously. Increasing n leads to a more stable estimate, but at the expense of increased sensitivity to EPS reference frame drifting.

5.1.2. Experimental Evaluation

In Figure 13 we see the test matrix used to evaluate our UPF catheter localization algorithm. Tests are classified using the three categories shown to the left: namely, which control input is used for the state transition model (EPS sensor or kinematics), what type of localization is performed (global or tracking), and what external conditions are applied to disturb the system.

Global localization is accomplished by imaging a swept volume of the phantom from a fixed position and honing in on

Control	Sensor-Based Estimation	Kinematics-Based Estimation	
Type	Global Localization	Global Localization	Tracking
Conditions	Stationary vs. Drifting Anatomy	Stationary vs. Drifting Anatomy	
	Stationary vs. Drifting EPS Reference		

Fig. 13. Evaluation test matrix. Testing is organized based by UPF control input, localization type, and disturbance condition.

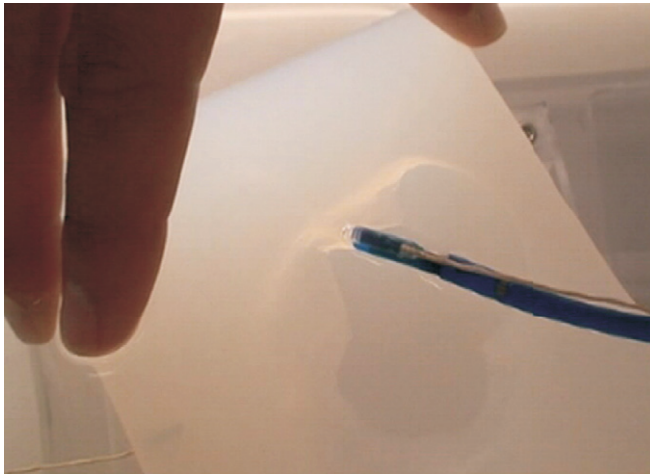


Fig. 14. Anatomy Drifting. We emulate drifting anatomy by physically moving the phantom by hand during navigation.

a pose estimate based on the acquired images. Tracking commences once we have converged upon a global pose estimate; we subsequently navigate inside the phantom and update our estimate each time step.

To test for disturbance rejection we emulate two drifting conditions: anatomy drifts, and EPS reference frame drifts. Anatomy drifts are emulated by moving the phantom by hand during imaging (Figure 14). We use the $^{EP}T_{meas}^{CT}$ sensor measurement to keep track of the exact amount of drift applied to the phantom. EPS reference drifts are emulated by adding a pre-computed drift sequence onto the $^{EP}T_{meas}^{IC}$ sensor measurements.

5.1.3. Estimate Error

At each time step we compute the position and orientation error of our UPF estimate to quantify the accuracy of our localization algorithm. When using EPS measurements as our control input, we take $^{CT}T^{EPk}$ to be our final estimate. When kinematics are employed, our final estimate is $^{CT}T^{ICk}$.

When using EPS sensors as our control input, position error is computed as the Euclidean distance between the estimated and measured coordinate origin offset:

$$error_{pos} = \|^{CT}O^{EPk} - ^{CT}O_{meas}^{EP}\|. \quad (46)$$

Orientation error is computed as the minimum angle required to transform estimated coordinate rotation offset $^{CT}R^{EPk}$ into measured offset $^{CT}R_{meas}^{EP}$ using an angle-axis rotation scheme:

$$error_{orient} = f_{angle}\left(\left(^{CT}R^{EPk}\right)\left(^{EP}R_{meas}^{CT}\right)\right). \quad (47)$$

When using the kinematic model as our control input, the position and orientation errors are similarly computed as

$$error_{pos} = \|^{CT}O^{ICk} - ^{CT}O_{meas}^{IC}\|, \quad (48)$$

$$error_{orient} = f_{angle}\left(\left(^{CT}R^{ICk}\right)\left(^{IC}R_{meas}^{CT}\right)\right). \quad (49)$$

5.2. Sensor-based Global Localization with Stationary EPS Reference

Global localization performance is tested by rotating the ICE catheter about its axis using a 100° amplitude, 1/10 Hz sine wave trajectory, while acquiring ICE images of the phantom at a penetration depth of 60 mm. This technique allows us to image a large volume of the phantom while moving the ICE catheter tip minimally. We initialize 500 UPF particles as described in Section 4.4 and estimate $^{CT}P^{ICk}$ over the course of four rotation cycles. After the first rotation cycle is complete, we compute a least-squares estimate of $^{CT}T^{EPk}$ at each time step for the remaining three rotation cycles. Each least-squares estimate incorporates all $^{CT}P^{ICk}$ estimates from the previous rotation cycle (the last $n = 150$ estimates). This test is repeated 50 times, each time using a different catheter configuration.

In Figure 15 we see the position and orientation estimate errors for each time step, averaged over the 50 trials. After the first rotation cycle is complete ($t = 0$), initial estimate errors are 5.36 mm and 30.15° . Over the next three rotation cycles, the estimate errors decrease to 0.79 mm and 2.37° .

5.2.1. Observed versus Expected Estimate Error

The observed estimate errors closely match the expected errors of our algorithm given the limitations of our experimental setup. As mentioned in Section 4.2, the position resolution

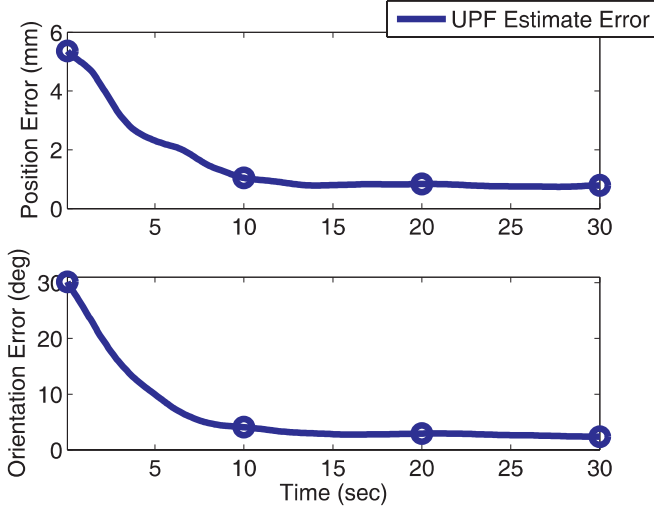


Fig. 15. Sensor-based global localization. We see the position and orientation errors of our UPF estimate over three rotation cycles. The termination of each rotation cycle is marked by a circle. We compare results using the original (warped) ICE images, and unwarped ICE images.

of our EPS sensor is 0.5 mm along each axis, resulting in an expected measurement error of approximately 0.433 mm. In Section 4.3.4 we determined that our scan-line distance and angle errors were approximately $r_{\text{err}} = 0.009$ mm/mm and $\vartheta_{\text{err}} = 0.001^\circ/^\circ$. When considering all of the pixels in an ICE image of 60 mm penetration depth, the average scan-line distance and angle are approximately $r = 40$ mm and $\vartheta = 22.5^\circ$. Using these values, we expect our calibration scheme to contribute around an additional 0.36 mm of error to our position estimate. Combined, the measurement and calibration errors add up to about 0.793 mm, which is roughly equivalent to our observed error (0.79 mm). The accuracy of our orientation estimate is limited by the elevation angle resolution of the ICE catheter ($\approx 2.5^\circ$), a byproduct of out-of-plane ultrasound beam spreading, described by Bushberg et al. (2002). The observed orientation error is consistent with this value.

5.2.2. Estimate Confidence

A convenient byproduct of our UPF estimation algorithm is that the state covariance matrix (Equation (43)) serves as a measurement of confidence in our estimate. In Figure 16 we plot the state covariance norm against the commanded rotation angle for each time step during one global localization trial. Over time, the covariance norm decreases as our estimate accuracy (and confidence) improves.

Within each rotation cycle, the covariance norm increases and decreases in accordance to the commanded rotation angle.

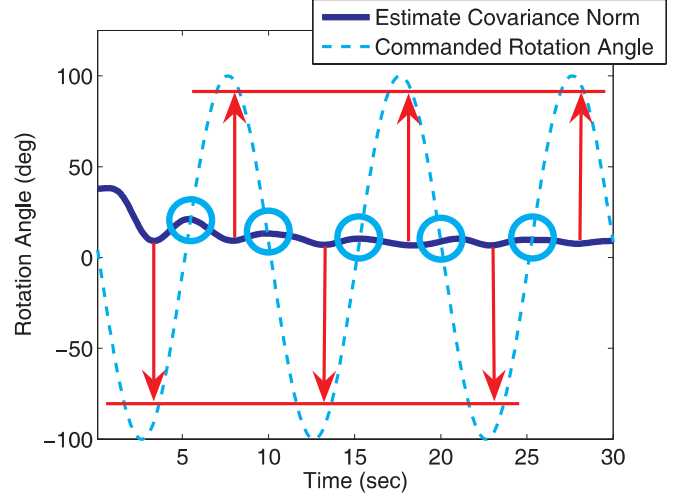


Fig. 16. State covariance and estimate confidence. As our estimate accuracy improves, our state covariance norm decreases. Therefore, the covariance norm can be taken as an estimate of confidence. During each rotation cycle our confidence decreases (circles) when we image the uniform posterior wall, and increases (arrows) when we image the pulmonary veins. As such, our covariance norm increases and decreases respectively during those time periods.

This behavior is expected because our confidence in estimating the correct state parameters is limited by the region of the anatomy we are currently imaging. On the left of Figure 17(a) we see an image of the mostly-uniform posterior wall taken at commanded rotation angle $\theta = 10^\circ$, and on the right we see a plot of correlation scores for virtual ICE images generated from nearby catheter configurations.

Conversely, on the left of Figure 17(b) we see an image of the feature-rich region surrounding the pulmonary veins taken at commanded rotation angle $\theta = 75^\circ$, and on the right we see a similar correlation plots for nearby configurations. The range of configurations generating correlation scores similar to that of the true configuration is larger when imaging the anatomic center, reducing our confidence in our estimate's accuracy. This is manifested as an increase in the state covariance norm.

5.3. Sensor-based Global Localization with Drifting EPS Reference Frame

Next, we evaluate the ability of our sensor-based global localization algorithm to compensate for an artificial drift introduced into the EPS measurements. The experimental setup and estimate computations are identical to those described in Section 5.2. As before, we compute the estimate error for each

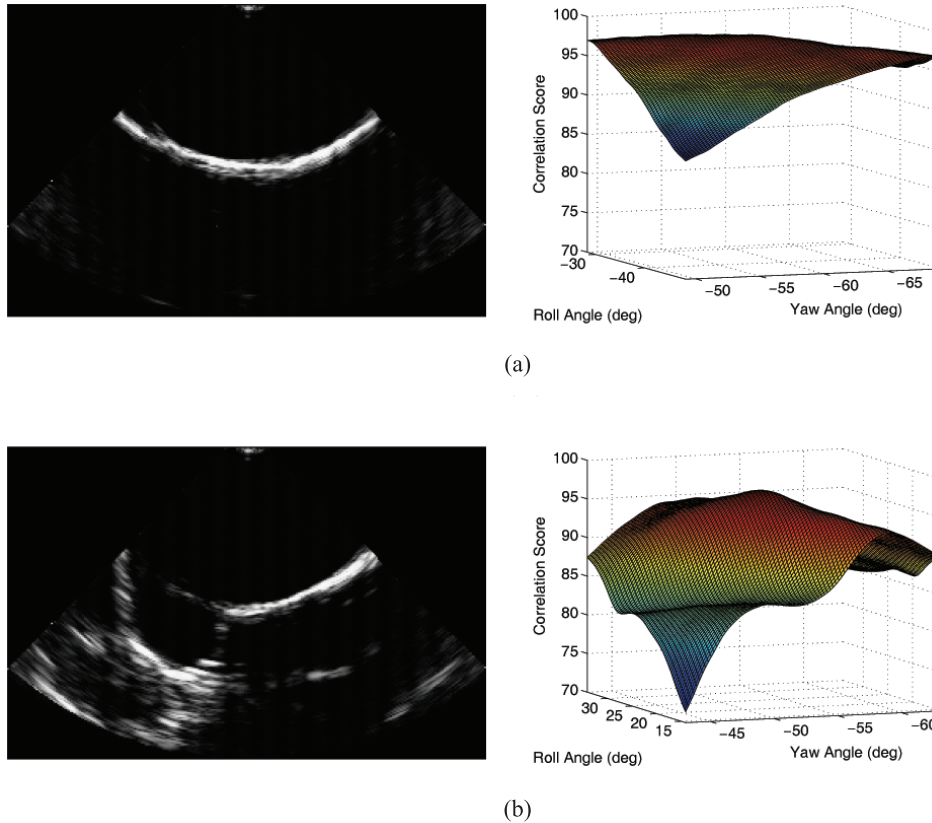


Fig. 17. Region-based estimate confidence. Depending upon which region of the anatomy is imaged, the range of configurations yielding virtual ICE images with similar correlation scores can vary greatly. In (a) we see an image of the posterior wall taken using roll and yaw angles $\alpha = -38^\circ$ and $\gamma = -59^\circ$, while in (b) we see an image of the pulmonary veins taken using roll and yaw angles $\alpha = 24^\circ$ and $\gamma = -53^\circ$. The steepness of the dropoff in virtual ICE correlation scores for configurations neighboring the actual catheter pose reveals that the range of similar configurations is larger when imaging the posterior wall and smaller when imaging the pulmonary veins.

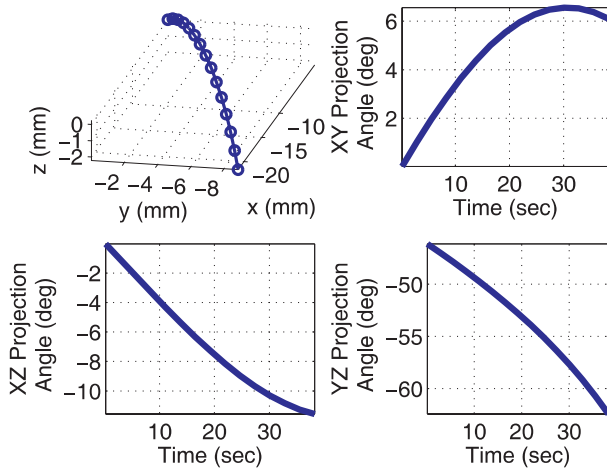
time step after the first rotation cycle, and present results averaged over 50 trials using different catheter configurations and EPS drift profiles. Figure 18(a) demonstrates the drift added to EPS measurements in one particular trial. While the amount drift increases over time, the absolute magnitude does not affect our algorithm since we consider EPS measurements differentially.

In Figure 18(b) we see the performance of our UPF estimate in comparison to a naive, “static” estimation scheme, where ${}^{CT}T_k^{EM}$ is assumed constant over time. We initialize the static scheme’s estimate to ${}^{CT}T_{meas}^{EP_0}$ so that, at first, there is no error in its estimate. The UPF estimate error is initially 9.53 mm and 76.49° , but by the end of the fourth rotation cycle it has dropped to 1.11 mm and 2.56° , again roughly at our performance limits. The static estimate’s error, on the other hand, grows to 27.9 mm and 11.2° , mirroring the amount of EPS drift introduced.

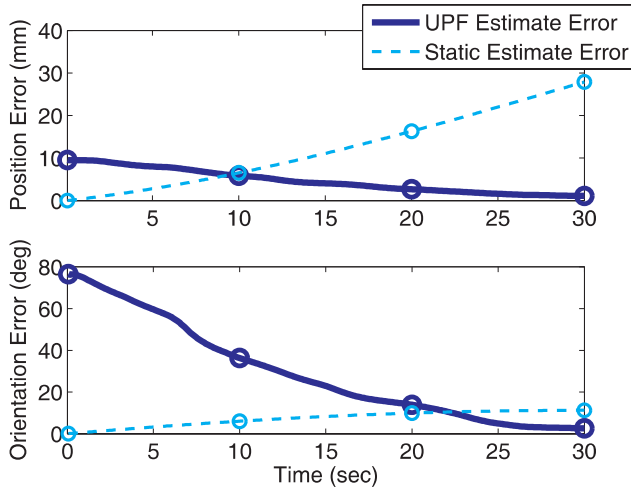
5.4. Kinematics-based Global Localization

We evaluate the accuracy of our kinematics-based global localization algorithm as described in Section 5.2, except here we estimate ${}^{CT}P^{IC_k}$ instead of ${}^{CT}T^{EP_k}$. In Figure 19 we compare the UPF estimate’s error to that of a kinematics-only estimation scheme. The kinematics-only estimate is initialized to the correct starting pose ${}^{CT}T_0^{IC}$, and uses the kinematic model to compute subsequent pose estimates.

After one rotation cycle ($t = 0$) our UPF estimate’s errors are 7.97 mm and 43.56° , and after four rotation cycles these errors have converged to 1.62 mm and 3.98° . The major reason for the increase in error (as compared with the results of our sensor-based localization algorithm) is that our performance is no longer limited by EPS sensor resolution, but instead by the resolution of our virtual ICE generation algorithm and NMI scoring system. When using kinematics as our control input,



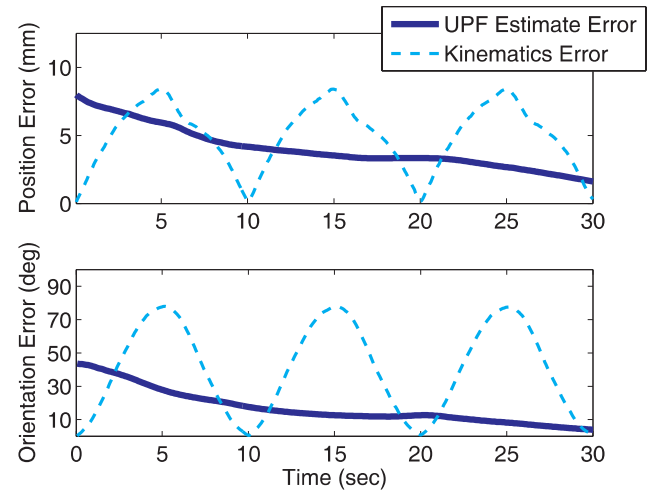
(a)



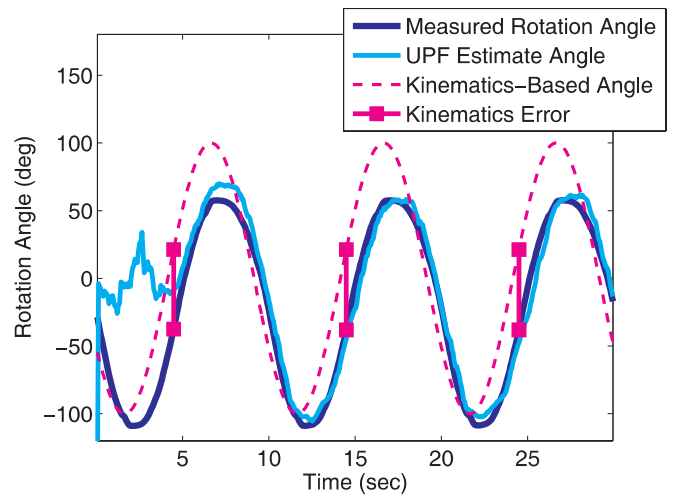
(b)

Fig. 18. Sensor-based global localization with drifting EPS. In (a) we see the EPS drift profile from one trial, where the position measurement (upper left) drifts 25 mm along a curved trajectory. The remaining three plots show the planar projection angles of the orientation drift over time. In (b) we see our UPF estimate results compared to a static estimation scheme that assumes no drift.

there is greater uncertainty in the amount of pose change at each time step. In Figures 17(a) and 17(b) we saw that a range of configurations surrounding the true catheter pose can yield virtual ICE images with similar correlation scores; as such, we have greater difficulty discerning how much the catheter actually moved on the basis of acquired ICE images alone. Furthermore, our ability to distinguish between configuration likelihoods is highly dependent on what region of the anatomy is imaged.



(a)



(b)

Fig. 19. Kinematics-based global localization. In (a) our UPF estimate errors are compared with a kinematics-only estimation scheme that assumes the robotic guide kinematics are correct. In (b) we see the measured, estimated, and commanded rotation angles from one particular trial. The estimated angle closely tracks the measured angle after the first rotation cycle, but the expected (commanded) rotation angle is off both in amplitude and phase.

Figure 19 also reveals that the kinematics-only estimate error increases and decreases depending upon the commanded rotation angle. In Figure 9(b) we saw that the actual amount of rotation observed at the ICE catheter tip depends upon the amount of distal bend applied to the robotic guide. In this particular experiment we applied $\psi = -30^\circ$ distal bend to the guide, and the actual rotation amplitude was around 80°

instead of 100° . In Figure 19(b) we see the measured, estimated, and commanded rotation trajectories from one localization trial. In addition to amplitude scaling, there is also a phase delay between the commanded and measured rotation trajectories.

In Figure 19(a) we see that the kinematics-only estimate goes off by 78° periodically in accordance with this amplitude and phase disagreement. The kinematics-only estimate also goes off by 8.42 mm at the same point in the rotation cycle; this is because the ICE catheter tip wobbles during its rotation, causing it to precess along an arc. The UPF estimate, on the other hand, is able to compensate for this behavior despite the relatively crude state transition noise computation used to compensate for kinematic model errors.

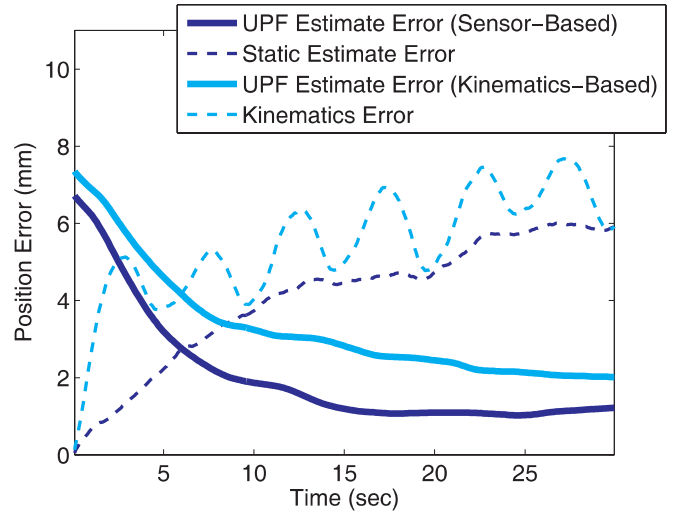
5.5. Sensor and Kinematics-based Global Localization with Drifting Anatomy

Next we evaluate the ability of our UPF algorithm to compensate for anatomic drift by manually translating and reorienting the phantom during imaging. As before, we present results averaged over 50 trials using different catheter configurations and phantom disturbance inputs. In Figure 20 we see our estimate errors when using a sensor-based control input, a kinematics-based control input, the static estimation scheme described in Section 5.3, and the kinematics-only estimation scheme described in Section 5.4.

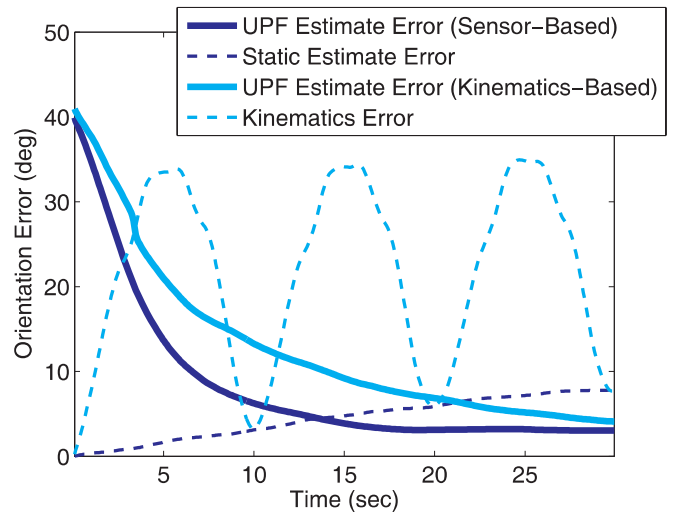
Despite the anatomic drift introduced during localization, both UPF estimates perform approximately as well as they did in the absence of disturbances. The sensor-based UPF estimate error ultimately drops to 1.22 mm and 3.04° , while the kinematics-based UPF estimate error drops to 2.01 mm and 4.11° . In both cases, the UPF estimate errors fall below the amount of drift applied to the phantom; the static estimate error, which mirrors the amount of phantom movement, grows to 5.9 mm and 7.87° by the end of the fourth rotation cycle. The kinematics-only estimate is affected by both kinematic model errors and phantom drift. Its errors increase and decrease periodically in accordance to the commanded rotation angle, while gradually increasing along with the amount of phantom drift.

5.5.1. Adaptive Resampling

As expected, when we encounter an input disturbance our algorithm's estimate confidence decreases, triggering an increase in the number of particles deployed as described in Section 4.4.2. In Figure 21 we see the average number of particles utilized by the sensor-based UPF estimate plotted against the average disturbance over time. The magnitude of the disturbance was computed as a function of the change in position and orientation of the phantom at each time step.



(a)



(b)

Fig. 20. Sensor and kinematics-based global localization with drifting anatomy. While moving the phantom by hand, we run global localization trials using both EPS measurements and kinematics as control inputs. In (a) we compare the position errors of the two UPF estimates to those of a static estimate that assumes no drift, and a kinematics-only estimate that assumes no kinematic deviations. In (b) we compare the orientation errors from the same four estimates.

We see that when the disturbance magnitude increases, the number of particles spikes, and when it decreases the number of particles settles back down. As disturbances were applied continuously, the number of particles gradually increase over time. However, this increase is modest compared with the number of particles initially deployed (500). In datasets where

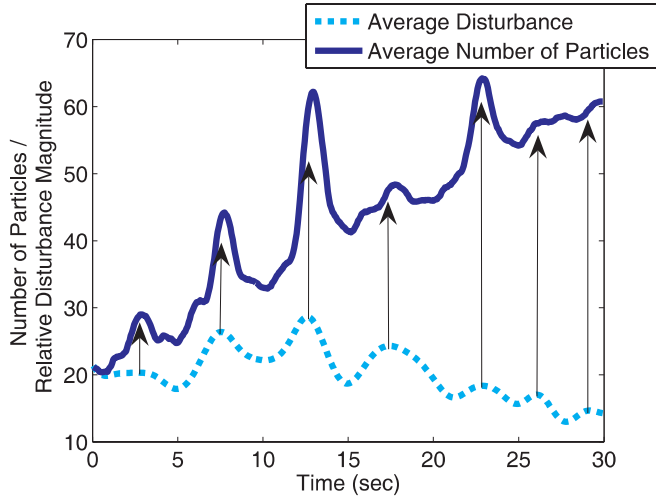


Fig. 21. Adaptive resampling. As disturbances are continuously applied to move the phantom, the number of particles used in the UPF estimate increases. When the magnitude of the disturbance spikes intermittently, so do the number of particles.

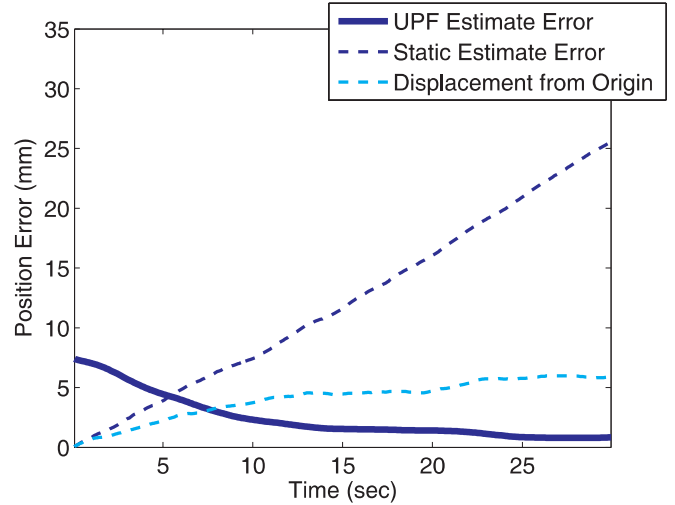
the anatomy either stops drifting or does not drift at all, the number of particles settles to a range between 25 and 50.

5.6. Sensor-based Global Localization with Drifting EPS Reference Frame and Anatomy

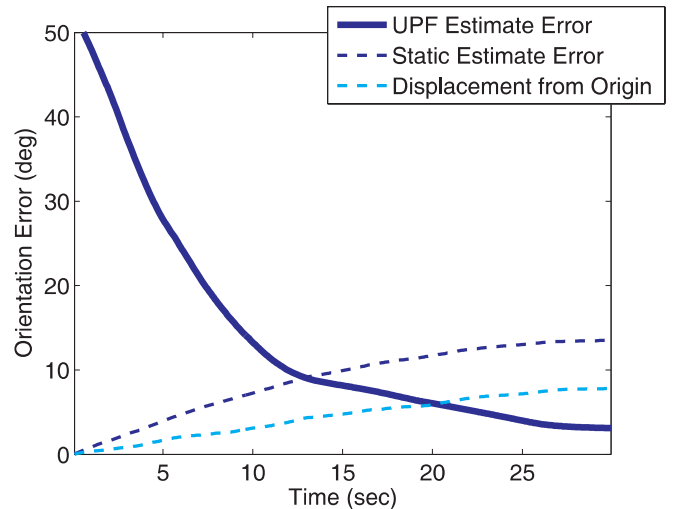
Finally, we evaluate the sensor-based global localization algorithm's ability to compensate for both EPS reference frame drifts and anatomy drifts concurrently. In Figure 22 we see the performance of our UPF estimate in comparison with the static estimation scheme, as well the average drift applied to the phantom by hand each trial (5.91 mm and 7.88° by the end of the fourth rotation cycle). The static estimate error is affected by both anatomy and EPS drifts, and gradually increases to 25.54 mm and 13.54° . The UPF estimate error, on the other hand, is unaffected by the drifts, and ultimately settles to 0.83 mm and 3.0° , once again at our limits of performance.

5.7. Kinematics-based Tracking

Once we have localized the ICE catheter with respect to the left atrium phantom, we begin driving the catheter inside the phantom using the robotic guide. We continue to acquire ICE images at 15 Hz and compute a new pose estimate after considering each image. Although the control input driving our UPF estimate can be EPS sensor-based, here we present results only from trials using the robotic guide's kinematics.



(a)



(b)

Fig. 22. Sensor-based global localization with drifting EPS reference frame and anatomy. While introducing drift into EPS measurements and moving the phantom by hand simultaneously, we ran a sensor-based global localization trial and compare our estimate's position error (a) and orientation error (b) to those of a static estimate that assumes no drift. As a reference, we also show the amount of drift applied to the phantom over time. Our UPF estimate is unaffected by both drifts.

In Figure 23(a) we see how the kinematics-based UPF estimate compares with the kinematics-only estimate as different joint commands are applied to the robotic guide. We see that the UPF estimate error stays below 7.5 mm and 13.5° throughout navigation, even as the kinematics-only estimate error grows to 21.2 mm and 38.7° . As the accuracy limitations

We did not evaluate our ability to globally localize the catheter when imaging only a portion of the left atrium; we felt it would be difficult to choose an unbiased test-case because of region-based sensitivity in our image correlation estimate confidence (Section 5.2.2). Depending upon which regions were imaged, we could have shown reasonable or poor performance, and it would have been difficult to make objective claims regarding our results.

The incremental nature of our localization algorithm allows us to “cancel out” error accumulation in both the EPS measurements and kinematic model. For example, when continuously rotating the ICE catheter from $\theta = \theta_0$ at $t = 0$ to $\theta = \theta_N$ at $t = N$, we assume that between any two time steps the EPS has a small amount of drift (${}^{EP}T_{\text{drift}_k}^{IC}$) introduced into its measurements, and the kinematic model has a small amount of error (${}^{RB}T_{\text{err}_k}^{IC}$) introduced into its pose estimate. At $t = N$, the EPS position estimate will have accumulated an error of magnitude $\|\sum_{k=0}^N {}^{EP}O_{\text{drift}_k}^{IC}\|$, while the kinematics position estimate will have accumulated an error of magnitude $\|\sum_{k=0}^N {}^{RB}O_{\text{err}_k}^{IC}\|$. As our localization estimate compensates for the error introduced at each time step, it will only accumulate errors of these magnitudes in the worst-case scenario.

With no prior knowledge, our algorithm accurately estimates the catheter’s pose after considering less than a minute’s worth of data. This is substantially faster than commercially available EPS mapping solutions, which can require more than an hour to build a comprehensive map of the anatomy. When using EPS measurements, our estimate falls to within approximately 1 mm of position and 2.5° of total orientation error. When using the kinematic model, our estimate falls to within approximately 2 mm of position and 4° of orientation error. Our algorithm is robust to both EPS measurement drift and anatomical movement, as emulated by moving the phantom away from its initial configuration during imaging.

After localizing the ICE catheter, we can compute registration parameters linking EPS measurements to atrium coordinates provided that an EPS sensor is mounted onto the ICE catheter. This registration allows us to compute where the ablation catheter lies inside the atrium.

If the ICE catheter is not sensorized, we can track its movement inside the atrium with input from the robotic guide’s kinematic model. Our algorithm is able to track the catheter pose’s better than an estimation scheme that uses the kinematic model alone, although its performance is limited by the “feature-richness” of the region of the anatomy being imaged. A solution to this limitation might be to consider the last n ICE images at each time step, increasing the overall feature-content under consideration. The downside of this strategy would be increased sensitivity to patient motion. However, assuming that these motions are “slow-moving”, we can counteract this sensitivity by considering only a few samples of the most recently acquired data.

In the future, we will address issues pertaining to anatomic deformations caused by the heart beat, as well as less-pronounced deformations that occur in the time between the CT/MR acquisition and the start of the ablation procedure. One solution to dealing with beating heart deformations is to ECG-gate ICE acquisitions, so that ICE images are only acquired during a particular phase of the heart beat cycle. Zhong et al. (2005) proposed a four-dimensional CT registration algorithm whereby a collection of CT datasets are acquired at different phases of the heart beat; during the procedure, ICE images would be compared with virtual images generated using the CT mesh model corresponding to the current heart beat phase. To compensate for less-pronounced deformations resulting in a morphological mismatch between the left atrium and the reference mesh model, we envision a scheme where ICE images are used not only for localization, but to modify and correct the reference mesh as well.

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